

Kovermann Family Statistical Analysis

A demographic, social, and geographic study
of 64,507 documented relatives

Andreas Karl Kovermann
65 sections · 1565–2024
Osnabrück · Minden-Lübbecke · Ohio · Texas
May 2026

Statistical Analysis — Kovermann Family Tree

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DATA COMPLETENESS

January shows +112% births and +106% deaths vs other months. This reflects persons for whom only the birth or death year is known — not a data error. FTM stores year-only dates as "1 JAN YYYY", a valid convention indicating the year is documented but month and day are not. Analyses requiring exact dates are filtered to fully-dated entries only. Roughly half of all persons have year-level but not day-level records.

1. LIFESPAN BY CENTURY

Century	Avg (all)	Avg (>5y)	Median	n
1500s	64.3y	64.3y	65y	89
1600s	55.7y	58.4y	61y	925
1700s	50.1y	55.9y	60y	4,211
1800s	54.6y	61.5y	67y	9,446
1900s	65.0y	67.7y	72y	10,278

The 18th century is the worst. The 1700s combine the Seven Years' War, Napoleon's early campaigns, chronic harvest failures, and maximum population pressure on the Heuerling system. The 16th century looks better partly because only documented nobles and wealthy burghers survive in the record.

Sex gap: Women outlive men by 2–7 years in every era. In the 1900s: men 61.9y, women 68.9y — the modern gap widens as male occupational and military risk increases.

2. DEATH YEAR SPIKES

Year	Deaths	Factor	Cause
2020	302	×1.47	COVID-19
1944	154	×1.48	WW2 peak
1918	126	×1.46	WW1 end + Spanish Flu
1811	104	×1.85	Napoleonic Continental Blockade + famine

1811 age/sex profile: M:49 F:54 — near-equal sex distribution. Largest group: infants 0–9 (37 deaths). No male excess in fighting ages. = civilian catastrophe signature (hunger, epidemic).

1812 pivot: M:46 F:18 — sudden male excess appears. Russian campaign begins. The demographic signal flips within a single year from civilian catastrophe to military mortality.

1918: Young male excess (20–30s peak) + Spanish Flu kills all ages. **1944:** 20–30s war deaths + heavy 60–80s civilian peak = bombing war.

3. CHILD MORTALITY

Era	Rate	Notes
1600s	8.4%	Under-documented
1700s	15.7%	Rising with population pressure
1800–1850	20.7%	Peak — 1 in 5 children died before age 5
1850–1900	16.6%	Slight improvement
1900–1950	7.9%	Modern medicine halves rate

346 families had 3 or more children dying before age 10. Worst case: Johann Caspar Eberhard Glüsing & Marie Elisabeth Düsterberg (~1782): 10 of 13 children died before age 10.

4. TWIN SURVIVAL

Era	Both survive	One dies	Both die
1700s	5%	9%	5%
1750s	16%	16%	22%
1800s	20%	13%	24%
1850s	30%	17%	20%
1900s	39%	8%	9%

608 same-day twin pairs identified. Pre-1850, both twins dying was more likely than both surviving. The 1900s cross the threshold where survival becomes the dominant outcome.

5. INTER-BIRTH INTERVALS

Century	Avg gap	Median
1600s	25.4 mo	26.2 mo
1700s	25.6 mo	26.1 mo
1800s	24.4 mo	24.6 mo
1900s	22.5 mo	22.2 mo

Remarkably stable across 400 years at ~2 years — breastfeeding as natural birth control. The 20th century drop reflects shorter nursing periods.

6. MARRIAGE AGE

Century	Men	Women	Gap
1500s	27.1y	22.6y	4.5y
1600s	29.5y	26.3y	3.2y
1700s	29.4y	26.5y	2.9y
1800s	29.4y	26.2y	3.2y
1900s	26.6y	24.0y	2.6y
2000s	29.4y	27.2y	2.2y

The 20th century marries earliest — welfare state enables it. The 21st century returns to 17th century marriage ages. The husband/wife gap narrows from 4.5y to 2.2y over 500 years.

7. HUSBAND/WIFE AGE DIFFERENCE

Century	H older by	H older	Same ($\pm 2y$)	W older
1600s	+3.8y	56%	31%	13%
1700s	+2.8y	49%	30%	20%
1800s	+3.4y	55%	31%	14%
1900s	+2.4y	42%	52%	6%
2000s	+1.7y	33%	60%	7%

Gap shrinks from 3.8 to 1.7 years over 400 years. Same-age marriages rise from 31% to 60% in the modern era. Wife-older cases (6–7%) are predominantly remarriages after widowhood.

8. REMARRIAGE RATES

Century	1 spouse	2 spouses	3+
1600s	86.0%	8.7%	5.3%
1700s	85.8%	9.3%	4.9%
1800s	90.7%	7.3%	2.0%
1900s	87.4%	11.4%	1.2%

The 1800s have the lowest remarriage rate — women raising children alone rather than remarrying. The 1900s rise to 11.4% second marriages includes both widowhood and rising divorce rates.

9. DIVORCE RATE

Era	Rate
1800s	~0%
1900s	9.9%
2000s	5.9%

Virtually zero before the 20th century. The 2000s figure is lower than the 1900s because many modern marriages are not yet concluded.

10. MARRIAGE DURATION

Century	Avg	Median	<10y	25y+
1600s	26.7y	26y	13%	54%
1700s	25.4y	25y	15%	51%
1800s	29.1y	29y	14%	60%
1900s	39.8y	42y	5%	81%

Marriage duration increases by 13 years from 17th to 20th century — direct result of rising life expectancy. 81% of 20th century marriages last 25+ years.

11. WIDOWHOOD DURATION

Century	Widows (F survives H)	Widowers (M survives W)
1600s	18.0y	17.1y
1700s	17.7y	15.5y
1800s	17.6y	17.0y
1900s	17.6y	13.1y

Widows survive husbands by ~17.6 years in every century — remarkably constant. But in the 1900s widowers survive wives for only 13.1 years, 4.5 years less — the modern female health advantage fully apparent.

12. ENDOGAMY — CLOSED MARRIAGE CIRCLES

Parish boundaries from Matricula Online (Bistum Osnabrück). Bauerschaften are NOT the unit — a Kirchspiel encompasses many Bauerschaften. Mentrup, Beckerode, Gellenbeck, Natrup Hagen are all St. Martinus Hagen.

Parish-level endogamy: ~73–76%, constant across all centuries.

Century	Same Kirchspiel	%
1600s	73.3%	stable
1700s	75.6%	stable
1800s	72.5%	stable
1900s	74.4%	stable

Beyond single parishes, three closed marriage circles exist:

Circle 1b — Hagen / Oesede / Osnabrück / Belm / Wallenhorst: 98.7% circle-endogamous 4,463 marriages internal, only 60 external. Internal flows: Hagen ↔ Hagen (1,196×), Osnabrück ↔ Osnabrück (1,132×), Hagen ↔ Osnabrück (490×), Oesede ↔ Osnabrück (362×), Hagen ↔ Oesede (204×). The dominant axis: rural Hagen ↔ urban Osnabrück.

Circle 2 — Ostercappeln / Bohmte: 44.2% circle-endogamous More open — a transitional zone between Circle 1 and Stemwede/Lübbecke. 63 marriages cross to Osnabrück, 55 to Stemwede. The Räuwer/Redeker/Kovermann lines sit in this transition zone.

Circle 3 — Minden-Lübbecke (Stemwede / Lübbecke / Rahden): 98.6% circle-endogamous 899 marriages internal, only 13 external. Stemwede ↔ Stemwede (464×), Lübbecke ↔ Lübbecke (245×), Lübbecke ↔ Stemwede (183×). An entirely separate genealogical world, connected by ancestry but not by marriage exchange. Only 14 cross-marriages to the Osnabrück network.

Circle 4 — Putnam / Allen / Van Wert / Auglaize / Mercer OH: 89.3% circle-endogamous 864 marriages internal, 104 external (counting all other Ohio cities — Cleveland, Dayton, Cincinnati, Columbus — and Indiana as external). With a strict 5-county core definition the rate is 99.5%; the 89.3% figure is the conservative reading. Allen ↔ Putnam (166×) is the dominant axis. Delphos (365 persons, Helmkamp/Osting/Grone leading) functions as a Putnam satellite despite straddling Allen/Van Wert. The 5 Osnabrück ↔ Putnam marriages in 300 years confirm near-total separation of German and American networks after emigration.

Effective endogamy (parish OR same circle): 94.0% Only 255 marriages in 270 years cross genuine circle boundaries.

Temporal stability of circles: | Century | Circle-endogamous | |---|---| | 1600s | 99.1% | | 1700s | 95.7% | | 1800s | 92.7% | | 1900s | 93.9% |

Slight opening in the 19th century (industrialization) but no real erosion. The circles remain intact into the 20th century.

13. MARRIAGE MONTH — CATHOLIC CALENDAR SIGNATURE

Pre-1850: October (+130%) and November (+157%) dominate massively. March (-67%) and December (-63%) nearly empty. = Canon law: no weddings in Lent (March) or Advent (December). Couples concentrated in the last possible weeks before Advent.

Post-1950: June and August lead — modern summer weddings. March still slightly negative (weather, not canon). The secularization of marriage timing is complete.

1850–1950: Transitional — November peak holds, May/June gaining.

14. SEASON OF DEATH — DISAPPEARANCE OF WINTER PEAK

Era	Winter	Spring	Summer	Autumn
1700s	32.5%	29.6%	17.6%	20.4%
1800s	27.9%	27.6%	21.6%	22.9%
1900–1950	26.4%	25.4%	24.1%	24.1%
1950–2010	23.5%	26.2%	25.9%	24.3%

Winter killed 32.5% in the 1700s, summer only 17.6% — nearly double. By 1950–2010 all seasons within 3% of each other. One of the clearest single signals of medical progress: modern heating, antibiotics, and nutrition erase the winter mortality season entirely.

15. FIRST NAMES — AMERICANIZATION TIMESTAMP

Era	Dominant names
1800–1899	Maria, Heinrich, Johann, Elisabeth, Anna
1900–1949	Mary, Ann, Joseph, John
1950–1999	Marie, Ann, Michael, Joseph, Lynn, James

The switch from German to English first names happens in a single generation, precisely at the 1900 watershed. American-born children of 1890s immigrants carry anglicized names.

16. SURNAME SHIFT — FROM KOVERMANN TO RECKER

Era	#1 surname	#2	Notes
1750–1799	Tiemann	Kovermann	Pre-emigration
1800–1849	Kovermann	Hehemann	Peak Kovermann
1850–1899	Recker	Meyer	Ohio lineage takes over
1950–1999	Recker	Schroeder	Nearly 4× most common

17. GEOGRAPHIC MIGRATION CROSSOVER

USA births first exceed German births in the 1890s.

Decade born	Germany	USA
1820s	1,001	0
1840s	1,000	62
1890s	833	1,044
1920s	167	2,078
1930s	24	2,505

Chain migration cutoff: emigration rate drops sharply after 1880. By 1890 the chain is broken — the family reproduces in America.

18. GENERATION LENGTH (mother's age at first child)

Century	Age at 1st child	Span	Avg children
1600s	28.8y	6.9y	3.2
1700s	28.5y	7.0y	3.3
1800s	26.7y	7.8y	3.9
1900s	24.9y	4.5y	2.7

1800s show maximum family size (3.9 children) with longest span (7.8y). 20th century compresses and shortens family formation dramatically.

19. FAMILY SIZE DISTRIBUTION

Era	Avg	8+ children
1650–1799	3.2	7–9%
1800–1849	3.5	10%
1850–1899	4.2	17%
1900–1949	3.4	7%
1950–1999	2.3	1%

The 1850s generation is the most fertile — 17% of families have 8+ children.

20. FIRSTBORN vs LASTBORN SURVIVAL

Era	Firstborn $\dagger < 5$	Lastborn $\dagger < 5$
1700–1749	14.7%	15.4%
1750–1799	24.4%	16.7%
1800–1849	21.8%	24.5%
1850–1899	15.3%	18.5%
1900–1949	7.1%	9.2%

Lastborn children die young more consistently — older, more exhausted mothers, larger resource-constrained families.

21. DEATH AGE DISTRIBUTION — FROM BIMODAL TO UNIMODAL

1750–1850: Strong bimodality — large infant peak at 0–4 (17.4%), deep valley at 5–14, rising to maximum at 65–74. Survival past age 5 predicts a reasonable lifespan.

1850–1950: Infant peak shrinks to 10.1%, adult peak shifts to 75–84. Transition from bimodal to unimodal complete.

Implication: In premodern Europe death had two faces — infant death and old age. The valley between them (ages 5–50) was relatively safe for those who survived childhood.

22. LONGEVITY HEREDITY

Children of short-lived parents (<65y avg): 63.1 years Children of long-lived parents (≥75y avg): 71.2 years Difference: 8.1 years (across 24,094 individuals)

A significant hereditary longevity signal. Caveat: shared environment and social class confound the genetic signal.

23. GEOGRAPHIC MOBILITY WITHIN GERMANY

- **54%** die in their birth Bauerschaft
- **46%** move to a different Bauerschaft

Within-parish mobility (Hagen a.T.W.): 83% cross Bauerschaft within the same Kirchspiel — explains the dense Beckerode ↔ Mentrup patterns which are intra-parish moves, not exogamy.

Mettingen: 100% same Bauerschaft — the most closed community. Consistent with the Brenninkmeyer clan's centuries-long concentration.

24. WAR AGE PROFILES

WW1 (181 male deaths): Peak 20–24y (50 deaths, 28%). Average age at death: 34.1y.

WW2 (220 male deaths): Peak 25–29y (41 deaths) but broad distribution across all ages including 55–64y (23 deaths). Average age at death: 36.5y — higher because civilian deaths inflate it.

KEY THRESHOLDS TIMELINE

Year/Period	Event
1700s	Life expectancy trough — worst century
1811	Civilian spike — blockade/famine (sex-equal deaths)
1812	Military spike — Russian campaign (male excess appears)
~1840–1860	Peak emigration decade
1800–1850	Peak child mortality (20.7%)
1890s	USA births first exceed German births
1900	First names switch German → English
1900–1950	Child mortality halved (20% → 8%)
1918	WW1 + Spanish Flu double spike
1944	WW2 peak mortality
~1950	Winter death season disappears
2020	COVID-19 largest modern mortality spike (×1.47)

25. COHORT SURVIVAL CURVES

Cohort	Survive to 5	to 40	to 60	to 80
1750–1799	86.9%	72.5%	47.7%	7.0%
1800–1849	81.0%	62.6%	42.0%	9.7%
1850–1899	85.3%	72.0%	59.3%	25.1%
1900–1949	93.0%	86.2%	75.1%	37.2%

The most dramatic single shift: surviving to age 80 rises from 7% to 37% across four cohorts — a fivefold increase. Surviving past age 5 rises from 81% to 93%: 120 per 1,000 children who died young in 1800–1849 would have survived in the 1900–1949 cohort.

Premodern death had two faces — a large infant peak and a broad adult plateau. Whoever survived past age 5 had a reasonable chance of old age. By 1900–1949 this bimodality disappears: death becomes concentrated in old age.

26. SIBLING DEATH CLUSTERS — EPIDEMIC YEARS

158 families with 2+ young children dying in the same year.

Epidemic year signatures (families with multi-child death):

Year	Families	Likely cause
1903	5	Scarlet fever / diphtheria
1855	4	Cholera
1871	4	Franco-Prussian War + epidemic sequelae
1811	4	Famine / Continental Blockade
1812	3	Russian campaign / famine
1886/1887	3 each	Childhood epidemics
1823	3	Typhoid/regional epidemic

1903 is the single worst year for sibling deaths — consistent with the scarlet fever / diphtheria waves in the Osnabrück region. Modern clusters (1979–1992) involve adult children: accidents and disease, not epidemics.

27. MARRIAGE DISTANCE — CONSTANT ACROSS 350 YEARS

Era	Avg distance	Median (all)	<5 km
1650s	3.0 km	1.8 km	74%
1750s	3.6 km	1.8 km	69%
1850s	4.0 km	2.0 km	70%
1900s	3.6 km	1.8 km	74%

The median marriage distance is 1.8 km in every century — including same-Bauerschaft marriages (distance 0). Among couples born in *different* places the median is 4.9 km (avg 6.3 km). Either way: no trend toward expansion despite industrialization. The Kirchspiel radius (~5 km) is the natural marriage market; 70–74% of all marriages fall within it. Cross-Kirchspiel marriages at 5–15 km represent 24–31% — consistent with the ~25% exogamy rate.

28. EMIGRATION SELECTIVITY

	Emigrants (†Ohio/USA)	Stayers (†Germany)
n	615	4,287
Avg birth year	1841	1836
% male	61%	51%
Avg lifespan	70.7y	60.2y

Male excess: 61% of emigrants are male vs. 51% of stayers. Classic chain migration: sons go first, earn, bring the family.

Lifespan paradox: Emigrants live 10.5 years longer than stayers. This is not America making people healthier — it is selection bias. Emigrants must be young, fit, and solvent enough to emigrate. The sick, elderly, and destitute stay. The "healthy emigrant effect" is clearly measurable in this dataset.

29. PARISH FERTILITY RATES

Parish	1800s avg children	1900s avg children
Glandorf	6.17	5.05
Mettingen	5.19	—
Putnam OH	5.52	3.99
Hagen	3.78	3.02
Osnabrück	3.93	2.71

Rural, strictly Catholic parishes (Glandorf, Mettingen) have the highest fertility — nearly double Osnabrück city in the 1800s. Putnam County Ohio (5.52 in 1800s) reflects abundant land enabling large families. The 20th century decline is universal but Putnam County remains well above the German comparison group at 3.99.

30. BIRTH/DEATH DAY OF MONTH

Days 1 and 31 are artefacts: - Day 1: systematic placeholder for month-unknown dates (filtered) - Day 31: underrepresented because not all months have 31 days (-42%)

Days 2–30 show no significant pattern — no end-of-month peak, no mid-month trough. Death and birth are randomly distributed within the month, consistent with no systematic recording bias for specific dates.

31. SAME-SURNAME MARRIAGES (cousin marriage proxy)

242 marriages between persons sharing a surname. Strong 20th century increase (115 cases 1900–1999) driven by Ohio community concentration — Meyer and Schroeder (16× and 15×) are both very common Putnam County names where co-surname does not necessarily imply close kinship. German same-surname marriages (pre-1900): only 15 cases — consistent with low cousin marriage rates in the Kirchspiel system.

32. GEOGRAPHIC CENTER OF GRAVITY

Year	Location
1750–1800	52.25°N 8.04°E — Osnabrück region
1825	52.18°N 7.44°E — Osnabrück (slight westward drift)
1850	50.92°N -2.89°E
1875	48.53°N -23.02°E
1900	45.83°N -45.89°E
1925	42.02°N -78.18°E — Pennsylvania/Ohio border
1950–2000	41°N -84°E

The family's geographic center of gravity crosses the Atlantic between 1875 and 1925 — visible as a smooth mathematical arc from Osnabrück to Ohio as the American population outgrows the German one. From 1950 the center stabilizes near Putnam County and barely moves.

33. BIRTH ORDER AND LIFESPAN

Position	Avg lifespan (adult deaths only)
Firstborn	67.3y
Middle children	68.8y
Lastborn	66.4y

Middle children live longest — 1.5y more than firstborn, 2.4y more than lastborn. Lastborn face older/more exhausted mothers and more resource competition. Middle children may benefit from established family resources without bearing the firstborn risk.

34. RURAL vs URBAN LIFESPAN

Era	Rural	Urban	Urban advantage
1750–1850	61.3y	59.4y	-1.9y
1850–1950	72.3y	68.1y	-4.2y
1950–2010	48.8y	50.1y	+1.3y

Cities killed until 1950. The urban penalty peaks at -4.2 years in the industrial era (overcrowding, pollution, epidemic density). Only after 1950 does urban infrastructure and medicine reverse this — for the first time in recorded history, city dwellers outlive rural populations.

35. OHIO SETTLEMENT WAVES

First documented births by decade: - **1835–1837**: Minster, Glandorf, Fort Jennings, Ottawa — the founding generation, arriving with or shortly after the Pennsylvania (1838 ship manifest) - **1843**: Cincinnati — urban parallel track - **1850s**: Putnam County, Mercer County, Maria Stein expanding - **1860–1890s**: Broad geographic diffusion across NW Ohio - **1920s**: Last new places settled — Bowling Green, St. Marys

Peak emigration by birth year: *1820–1824 (14 persons), emigrating ~1845–1849 — directly into the potato famine migration wave.

36. MARRIAGE CIRCLES — COMPLETE STRUCTURE

Based on Kirchspiel boundaries from Matricula Online (Bistum Osnabrück) and Ohio county records.

Circle 1b — Hagen/Oesede/Osnabrück/Belm/Wallenhorst/Schledehausen Internal: 4,463 marriages | External: 60 | **Endogamy: 98.7%**

The largest and most closed circle. Wallenhorst connects almost exclusively to Osnabrück (193×) and internally (72×). Belm is topographically and canonically part of this circle. Adding these three parishes raises endogamy from 96.0% to 98.7%.

Circle 3 — Minden-Lübbecke (Stemwede/Lübbecke/Rahden): 98.6% Stemwede ↔ Stemwede (464×), Lübbecke ↔ Lübbecke (245×), Lübbecke ↔ Stemwede (183×). Almost as closed as Circle 1b. Only 14 marriages cross to the Osnabrück network in total. A completely separate genealogical world within the same family tree

— connected by ancestry but not by marriage exchange. The Stemwede/ Lübbecke lines diverged from the Osnabrück core so early and so completely that the two networks barely interact.

Circle 2 — Ostercappeln/Bohmte: 44.4% endogamous A transitional zone, not a true closed circle. Connected to Circle 1 via Osnabrück (70×) and to Stemwede/Lübbecke (55×). Sits between two strong circles without fully belonging to either. The Räuwer/Redeker/Kovermann lines occupy this transition zone.

Circle 3 — Minden-Lübbecke (Stemwede/Lübbecke/Rahden): 98.6% Stemwede ↔ Stemwede (464×), Lübbecke ↔ Lübbecke (245×), Lübbecke ↔ Stemwede (183×). Almost as closed as Circle 1b. Only 14 marriages cross to the Osnabrück network in total. A completely separate genealogical world within the same family tree, connected by ancestry but not by marriage exchange.

Circle 4 — Putnam/Allen/Van Wert/Auglaize/Mercer OH: 89.3% More open than the German circles — Ohio is a mobile society. Key connections: Allen ↔ Putnam (166×), the dominant axis. Delphos (365 persons) acts as bridge between Allen and Van Wert counties: 64 marriages toward Putnam, 55 toward Allen, only 21 internal Van Wert. Delphos functions genealogically as a Putnam County satellite despite straddling the Allen/Van Wert border.

Effective endogamy (all German circles combined): ~98% The German circles are essentially isolated from each other except through Osnabrück as bridge. German and American networks are nearly fully disconnected: only 5 Osnabrück ↔ Putnam marriages in 300 years.

Bridge parishes: - **Osnabrück (74×)** — only bridge between German circles - **Ostercappeln (63×)** — connects Circle 2 to Circle 1 - Stemwede (9×) and Lübbecke (5×) — minimal bridges to Circle 3

37. SURNAME CONCENTRATION — GINI COEFFICIENT

Era	Gini	Unique surnames	Top-5 share
1700–1750	0.589	1,372	8.5%
1800–1850	0.661	3,064	10.3%
1850–1900	0.614	5,524	5.3%
1900–1950	0.527	11,485	4.5%
1950–2000	0.561	16,691	4.1%

Peak concentration in the 1800s when few families dominate. The Ohio immigration explosion after 1850 brings hundreds of new surnames, democratizing the name landscape. By 1950 the top-5 account for only 4.1% vs. 10.3% at the peak.

38. NAME TRANSMISSION

	Match parent's first name
Sons with father's name	12.6% (4,143/32,810)
Daughters with mother's name	9.3% (2,886/31,008)

Lower than expected. In the Catholic Osnabrück region, naming after godparents was more common than after parents. The "eldest son = father's name" pattern was not a rigid rule. (Underestimated — middle names, where the parental name often appears, are not captured here.)

39. PREMARITAL CONCEPTION (first child <9 months after marriage)

Century	Premarital conception rate
1600s	22.9%
1700s	18.5%
1800s	17.3%
1900s	16.4%

Nearly one in five first children was conceived before the wedding — constant across four centuries. This contradicts the image of a prudish past: premarital sexuality was socially normal as long as marriage followed. In rural society, pregnancy often triggered or confirmed the intent to marry — one married once fertility was proven.

40. MATERNAL MORTALITY

Century	Deaths per 1,000 births (mother died same year)
1600s	12.6
1700s	6.3
1800s	7.9
1900s	1.8

An 85% decline from the 17th to 20th century. The slight 19th century rise over the 18th may reflect childbed fever in early hospitals before Semmelweis's handwashing took hold. The 20th century collapse is antisepsis, antibiotics, and modern obstetrics.

41. AGE AT LAST CHILD — family planning revolution

Century	Avg age at last child	Median
1600s	36.1y	37y
1700s	35.7y	37y
1800s	34.6y	35y
1900s	29.6y	29y

Until 1800 women bore children into their mid-30s — up to the natural fertility limit. In the 20th century this drops to 29.6 — for the first time, couples voluntarily stop. The statistical fingerprint of deliberate family planning: contraception, smaller desired families. The median falls 8 years (37→29).

42. NEVER-MARRIED RATE (adults who reached age 25)

Century	Never married
1600s	13.3%
1700s	14.5%
1800s	25.8%
1900s	21.1%

The jump from 14.5% to 25.8% in the 19th century is striking — a quarter of all adults never married. This fits the emigration era and the Heuerling problem: those who could not inherit land and did not emigrate often remained single, as marriage required an economic foundation many lacked. Unmarried siblings stayed on the farm as labor. (Partly a data artefact — emigrants whose American marriages are unrecorded.)

43. CONCEPTION SEASONALITY (back-calculated from birth month)

Peak	Trough
June (+5.7%)	September (-7.8%)
Nov–Jan (+4%)	May (-4%)

Conceptions cluster in early summer (May–June) and late autumn/winter (Nov–Jan). The September trough is the strongest signal — likely exhaustion from harvest labor, when the entire rural population worked the fields dawn to dusk. The winter peak fits the dark season with little work and long nights.

44. ORPHANHOOD

- **12.1%** lost their father before age 15
- **12.5%** lost their mother before age 15
- Median age at father's death: 35; at mother's death: 38

About one in eight children was a half-orphan before puberty. Most people, however, were already adult when parents died. This explains the high number of step-parents and the low remarriage threshold for young widows with small children.

45. SURNAME GEOGRAPHIC ANCHORING

Surname	Core locations
Kovermann	Osnabrück (20), Oesede (17), Potthoffs Kotten (15)
Recker	Ohio (303), Glandorf (83), Ottawa (65)
Helmkamp	Delphos (27), Indiana (20)
Hallmann	Bohmte (37), Ostercappeln (18)

Each principal surname has a geographic core, and the cores map exactly onto the marriage circles: Recker = Ohio circle (4), Kovermann = Osnabrück circle (1), Hallmann = Ostercappeln/Bohmte circle (2). The surname geography confirms the circle structure independently.

46. DESCENDANT EXPLOSION — most prolific founders

Founder	Documented descendants
Dorothea Adelheid Brockschmidt (*1681)	25,898
Johann Heinrich Recker gen Redeker (*1710)	25,898
Johann Heinrich Recker gen Redeker (*1766, emigrant)	25,888
Margarethe Brünemann / Eberhard Forstmann (*1570)	25,690
GerdT Tiemann (*1557)	21,997
Jacob Ühmann (*1525)	20,785

The top founders all belong to the Recker/Redeker line that emigrated to Ohio and multiplied explosively. The oldest roots (Ühmann 1525, Tiemann 1557, Brünemann/Forstmann *1570) have 20,000+ descendants because they sit at the base of the entire Osnabrück network.

47. PEDIGREE COLLAPSE — the central ancestral bottleneck

28% of all 64,507 relatives (17,876) connect through a single common ancestor: Johann Heinrich Recker gen Redeker (emigrant, 1838).

This is the father of Gerhard Heinrich from the central descent case — the man whose family took the Pennsylvania in 1838. More than a quarter of all documented relatives connect to Andreas through him. The DNA logic of the 21 Putnam County matches becomes statistically concrete: this one man is the convergence point of a third of the entire tree.

Next-largest convergence nodes: | Ancestor | Relatives connecting through | |---|---| | Johann Heinrich Recker gen Redeker (1838) | 17,876 | | Gerhard Heinrich Redeker + Anna Maria Heuer | 5,685 each | | Heinrich Wortmann + Anna Magdalena Ühmann (~1631) | ~4,800 | | Conrad Friedrich Covers (illeg.) | 3,424 | | Johann Gerhard Bucker gen Hemesath | 2,119 |

48. RELATIONSHIP STRUCTURE

Relationship type	Count	Share
Cousin (Nth degree)	62,052	96.2%
Nth great-grand-uncle/aunt	1,358	2.1%
Nth great-grandparent (direct ascent)	1,063	1.6%
Other (siblings, direct)	~30	<0.1%

96% of the tree is "Nth-degree cousin" — the tree is overwhelmingly horizontal, built from distant collateral lines converging on shared ancestors several generations back. The 1,063 direct ancestors and 1,358 ancestral siblings form the vertical spine.

49. MARRIAGE WEEKDAY — the sharpest cultural shift

Pre-1900: Tuesday (22%), Wednesday (21.7%), Sunday (21.2%) lead. Saturday only 5.9%.

Post-1950: Saturday 72% — total dominance. Tue/Wed under 3%.

The single most radical behavioral change in the entire tree. In the Catholic 19th century one married on weekdays — Tuesday and Wednesday preferred, since Sunday belonged to worship, Friday (Christ's death) and Saturday (Marian day) were avoided. The 20th century inverts everything: the work-free Saturday becomes the only conceivable wedding day. The 72% Saturday share is the fingerprint of the modern work week.

50. SEX RATIO AT BIRTH — biologically normal

Era	M per 100 F
1700–1750	101.4
1800–1850	107.7
1900–1950	104.7
1950–2000	106.6

The natural ratio is ~105 boys per 100 girls. The tree matches almost perfectly — slight under-recording of girls in the 18th century (101) evens out later. Confirms data quality: no systematic sex bias.

51. REMARRIAGE AGE PATTERN

For persons with multiple marriages: - First spouse: avg age 25.0 at marriage - Last spouse: avg age 36.0 at marriage - Late spouses are +11.0 years older (n=1,647)

Counter-intuitive — one might expect widowers to marry younger partners. Instead the remarriage market sorts by age: a 36-year-old widow marries a widower of similar age, not a young first-time groom.

52. OLDEST VERIFIED INDIVIDUALS

Age	Person	Years	Place
114	Lieselotte Rustmeier	1910–2024	—
111	Catharina Maria Wulfetange	1776–1887	Wallenhorst
110	Hermann Heinrich Hemesath	1795–1905	(Heuerhaus)
110	Franz Henrich Schwalenberg	1800–1910	Wallenhorst
107	Maria Agnes Duhling	1811–1918	Wallenhorst

86 centenarians total. **Wallenhorst** appears repeatedly among the oldest (111, 110, 107 years) — a longevity cluster in a single parish, possibly genetic, possibly a well-documented community, possibly both.

CLOSING NOTE — THE TREE'S CENTER OF GRAVITY

The single most important structural fact: **one man, Johann Heinrich Recker gen Redeker, connects 28% of the entire documented family.** His son Gerhard emigrated in 1838; his granddaughter Anna Maria Elisabeth stayed behind and became Andreas's 2× great-grandmother through the illegitimate line. The genealogical, statistical, and DNA evidence all converge on this one bottleneck — the hinge between the German past and the American present of the family.

53. GENERATION GAP — constant across 1,000 years

Andreas's direct line reaches back to Generation 28 (~1003 AD). The gap between generations is remarkably stable at 30–35 years throughout — no systematic change from medieval to modern times.

Generation	Avg birth year	Gap
Gen 5	1824	30y
Gen 10	1656	35y
Gen 15	1472	—
Gen 20	1313	33y
Gen 25	1174	—
Gen 28	1003	—

A generation lasts a third of a century whether in the Middle Ages or the modern era. (Outliers in the oldest generations are statistical noise from very small samples, n=1–6.)

54. FARM NAMES ("genannt")

345 persons (0.5%) carry a farm name ("genannt"). Most common:

Farm name	Count
gen. Redeker	11
gen. Melcher	6
gen. Spreen	6
gen. Quade	5
gen. Meyer	5

"Genannt" names mark a person who took the name of the farm they married into or inherited — the Hofname overriding the birth surname. gen. Redeker leads, consistent with the central Recker/Redeker line.

55. TIME TO REMARRIAGE — widowers hurry, widows wait

	Avg time to remarriage	Within 1 year
Widowers	1.6 years	72%
Widows	2.5 years	57%

Widowers remarry nearly twice as fast. 72% within a year — economic necessity: a farmer urgently needed a wife for household, livestock, and children. A widow could take more time, often managing the inheritance or simply under less pressure. Remarriage speed directly measures the gendered division of labor.

56. CHILD COUNT vs MOTHER'S LIFESPAN

Children	Mother's death age (women who reached 40)
0	73.0y
1	68.9y
5	73.2y
10	74.4y
12	78.2y

Surprisingly, women with many children do not live shorter — if anything slightly longer. Mothers of 12 reach 78 on average. This is not evidence that childbearing is healthy — it is selection: only robust, healthy women could carry 12 children and reach 40+. The evolutionary "cost of reproduction" is masked by the selection effect.

57. TWIN RATE

18.1 twin births per 1,000 (962 twin births in 53,117 dated births, ~480 twin pairs). Higher than the historical natural rate (~12/1,000), likely a recording artefact — twins are reliably documented as a notable event while undated single births are underrepresented. True rate probably nearer 12–13/1,000.

58. FIRST-NAME DIVERSITY — the Simpson explosion

Era	Men (top name)	Women (top name)
1700–1750	0.694 (Johann 55%)	0.812 (Anna 36%)
1800–1850	0.865 (Johann 34%)	0.839 (Maria 31%)
1850–1900	0.967 (Johann 12%)	0.916 (Maria 22%)
1900–1950	0.988 (Robert 4%)	0.990 (Mary 6%)
1950–2000	0.988 (Michael 4%)	0.994 (Mary 3%)

The most dramatic shift is in men's names: in 1700, 55% of all boys were named Johann — an overwhelming monoculture. By 1950 the most common name is down to 4%. The Simpson index rises from 0.69 to 0.99 (near-maximal diversity). Women were always more diversely named (0.81 vs 0.69), but both converge to near-total diversity in the 20th century — the end of the saint-name monoculture and the start of individual name choice, exactly parallel to Americanization.

59. WINTER MORTALITY BY AGE (1700–1900, expected 25%)

Age	% winter deaths
0–9	28.7%
30–39	27.9%
50–59	30.2%
60–69	34.0%
70–79	34.9%
80–89	34.7%

Children and adults sit slightly above the 25% baseline (27–30%), but from age 60 the winter share jumps to 35%. The elderly die disproportionately in winter — cold, respiratory infection, cardiovascular strain. The classic "excess winter mortality" pattern, still present today but far more pronounced then. A 70-year-old faced a third higher risk of dying in winter than chance would predict.

60. WITNESS & GODPARENT NETWORK — social confirmation of the circles

Marriage witnesses (Trauzeugen) and baptism godparents (Paten) are recorded as free text in the event records. Extracting and matching against the real surname list reveals the social network beyond blood.

Families serving most often as marriage witnesses:

Family	Times as witness
Kovermann	53
Meyer	32
Witte	22
Bensmann	22
Menkhaus	21
Hallmann	19
Hemesath	17
Hune	16
Hehemann	14

Geographic origin of witnesses:

Kirchspiel	Share
Hagen	41%
Osnabrück	22%
Oesede	11%
Wallenhorst	10%
Ostercappeln	7%
Bohmte	6%
Belm	3%

The first four — Hagen, Osnabrück, Oesede, Wallenhorst — account for 84% of all witnesses, exactly the members of Circle 1b. Witnessing was not a break from the circle but its ritual confirmation: one witnessed the marriages of the same families one married into and stood godparent for. Kovermann leads at 53× — the surname sat at the center of the social web, not just the marriage network.

This is the third independent confirmation of the circle structure: first through marriages, then through surname geography, now through the witnessing network. Three distinct social mechanisms — marriage, name inheritance, ritual sponsorship — draw the same map.

61. WALLENHORST LONGEVITY CLUSTER — documentary, not genetic

Parish	Over 90	Over 100
Wallenhorst	4.0%	0.67%
Hagen	2.6%	0.11%
Osnabrück	2.5%	0.12%

Wallenhorst has a 6× higher centenarian rate than Hagen or Osnabrück. A statistically clear cluster — but the temporal distribution reveals its cause. Of 67 individuals aged 85+, 44 were born after 1875 (18 in 1875–99, 26 in 1900–24). Before 1850, only a handful.

A genetic cluster would distribute evenly across centuries. Instead it concentrates exactly where documentation is best and modern medicine operates. Decisive evidence: average lifespan in Wallenhorst before 1850 was actually *lower* than Hagen (55.4y vs 63.8y in 1850–99). The cluster is a documentation effect of the modern era: in a well-recorded parish the few centenarians are reliably captured, while elsewhere they fall through the gaps. The surnames of the elderly (Feldkamp 8×, Bartke 5×, Albers 5×) are too scattered for a true genetic signal.

Conclusion: documentary, not genetic.

62. FARM NAMES (Hofnamen) AND INHERITANCE — the marrying-in principle

375 persons carry a farm name ("genannt"). **354 (94%) are male** — the first strong signal: farm names are a male phenomenon.

The marrying-in hypothesis is confirmed: in 43% of men with farm names, the farm name exactly matches the *wife's* birth surname:

Man	Wife (née)
Peter Kreftsiek gen Leimkuhle	Leimkuhle
Johann Heinrich Kulgmeyer gen. Boberg	Boberg
Gerd Hermann Spreen gen Schnier	Schnier
Bernhard Friedrich Vincke gen Brockmeyer	Brockmeyer

This is the classic Westphalian Erbhof principle: when a man married onto a farm with no male heir, he took the farm name — "Kreftsiek genannt Leimkuhle" means born Kreftsiek but now master of the Leimkuhle farm. The farm mattered more than the bloodline. (The other 57% are cases where the farm name came from an earlier generation or the wife cannot be uniquely matched.)

Inheritance: 72% persistence — three of four children keep the farm name. Once a man took over the farm, the farm name became the family name of the next generation. The farm permanently "adopted" the family.

Temporal: Farm names peak in the 18th century (193 cases), nearly vanish after 1900 (7). This is the end of the *eigenbehörig* (bound-serf) farm system — as 19th century peasant emancipation converted farms to free property and industrialization loosened the tie to the land, the need to place the farm name above the bloodline disappeared.

gen. Redeker is anchored in Hitzhausen (6×) — exactly where Andreas's Redeker line originates. Direct evidence that "Redeker" itself was a farm name passed on through marriage, not a pure bloodline surname.

63. THE THREE EMIGRATION SYSTEMS — confession determined destination

The family did not emigrate to one place. Three separate, confessionally distinct emigration systems operated in parallel and did not mix. These clusters (Ohio, Texas, Pittsburgh — plus the Australian Thiemeyer/Haunert lines documented elsewhere) were already known; what this analysis adds is the confessional logic that explains the destinations, and the surname evidence for the direct village-to-village chains.

System 1 — Osnabrück/Glandorf → Ohio (Catholic)

411 documented emigrants. Origin: the Catholic Prince-Bishopric of Osnabrück (Hagen, Oesede, Glandorf, Ostercappeln). Destination: the German-Catholic parishes of northwest Ohio — Glandorf, Delphos, Lima, Ottawa, Maria Stein, St. Henry. Peak emigration birth cohorts: 1840s and 1890s. This is the main family line (Recker/Redeker/Kovermann).

System 2 — Minden-Lübbecke → Texas (Protestant)

93+ documented emigrants. Origin: Protestant Prussian Minden-Lübbecke (Stemwede, Wehdem, Lübbecke, Rahden). Destination: German-Protestant settlements in Austin/Washington County, Texas — Brenham and literally "**New Wehdem, Austin County, Texas**", a daughter settlement named after Wehdem. Peak: 1840s–1860s.

Direct migration chain confirmed: 23 surnames appear in both Wehdem (Germany) and Texas — Melcher, Holle, Spreen, Schomburg, Wendt, Winkelmann, Gaskamp, Quebe, Lehde, Bockhorn, Cassen, Grube, Piper, Rodenbeck and more. The New Wehdem families are demonstrably the emigrated Wehdem families.

System 3 — Pittsburgh (urban-industrial)

72 deaths in Pittsburgh. Families: Hune, Dittman, Lamkemeyer, Meyer, Hengelsberg, Werries — originating partly from Osnabrück/Harderberg. A smaller urban-industrial chain distinct from the rural Catholic Ohio and Protestant Texas systems. (A minor Iowa/Dubuque chain also exists, ~9 persons.)

The decisive factor: confession

- **Catholic** Osnabrück/Glandorf → Catholic Ohio parishes
- **Protestant** Minden-Lübbecke → Protestant Texas settlements

This is not coincidence. Religious affiliation determined the migration chain. Catholics followed Catholic chains to Ohio, Protestants followed Protestant chains to Texas. The two clusters never merged — just as Circle 1 (Osnabrück) and Circle 3 (Minden-Lübbecke) barely intermarried in Germany, their emigrant branches landed on opposite ends of America. The confessional boundary that separated the German marriage circles reproduced itself across the Atlantic.

Minden-Lübbecke cluster profile

2,579 persons born there (1565–2001), peaking 1700–1850. Top families: Meyer, Priesmeyer, Winkelmann, Spreen, Piper, Quade, Holle, Lehde, Krohne. Of those with recorded deaths: 1,320 stayed in Minden-Lübbecke, 243 went to the USA (overwhelmingly Texas), only 10 to Osnabrück — confirming the near-total separation from the Osnabrück network.

CLOSING — TWO FAMILIES, TWO WORLDS, ONE TREE

The deepest structural finding of the entire analysis: this is not one family that emigrated to one place, but a federation of confessionally and geographically distinct marriage circles, each of which transplanted itself to a different corner of America along its own religious chain.

- The **Catholic Osnabrück core** (Circles 1, 2) → **Ohio**
- The **Protestant Minden-Lübbecke world** (Circle 3) → **Texas**

They share Andreas as a descendant but barely shared a wedding in 300 years. The single bottleneck connecting 28% of the tree — Johann Heinrich Recker gen Redeker — belongs entirely to the Catholic Ohio system. The Minden-Lübbecke/Texas world is a parallel universe joined to the main tree only at its outer edges.

64. RESIDENCE DATA (RESI) — lifetime migration paths

The GEDCOM holds 197,971 residence entries for relatives (35,776 persons, avg 5.5 residences each, max 28). This is overwhelmingly American data: RESI-year peaks fall exactly on US census years (1950: 10,509; 1940; 1935; 1930; 1920; 1910) plus modern address databases (1993–2000). German records, by contrast, capture only birth-marriage-death.

Lifetime mobility — sedentariness collapses around 1900

Era	Avg distinct places/life	% sedentary (1 place)
1750–1799	1.37	65%
1800–1849	1.47	61%
1850–1899	2.21	46%
1900–1949	3.96	11%
1950–1999	3.60	12%

One of the sharpest behavioral shifts in the tree. Until 1850, two-thirds of all people lived their entire lives in one place. By the 20th century only 11% stayed put; the average person moved nearly four times. The break is precisely around 1900 — industrialization, railroads, and American internal migration. (Partly a documentation effect: US censuses record every move every decade, while German church books do not.)

Reconstructed migration paths — the American westward trek

The residence chains trace the great westward migration:

- **Mary Catharine Recker** (*1847 New Vienna): Dubuque → Dyersville → St. Helena → North Fork → Nebraska → †1924 Bow Valley. An East-coast- to-Great-Plains migration over 77 years.
- **Andrew Frank Recker** (*1877): Iowa → Kansas → El Paso → Oregon → California (Santa Ana). The Recker line from Iowa to the West Coast.

- **James Cornelius Drerup** (*1930 Glandorf): stayed entirely within the Putnam County core (Ottawa, Blanchard, Delphos, Lima) — the Catholic Ohio line remained locally anchored while the Iowa Recker moved west.

Transatlantic individual paths

- **Maria Catharina Grothaus** (*Harderberg) → Pittsburgh (1850, 1870), confirming the Pittsburgh chain from the southern Osnabrück area.
- **Heinrich Hellmich** (*Holzhausen) → six decades in Collinsville, Illinois (1900–1940) — a settled emigrant who never moved after arrival.
- **Heinrich Wilhelm Buck** (*Minden-Lübbecke) → Indianapolis/Perry, Indiana over 50 years — the Protestant line branched to Indiana, not only Texas, showing sub-branches within each confessional system.

The residence data confirms the broader pattern: the German past was sedentary (one place per life), the American present is mobile (4 places per life), and the transition is the emigration itself — the act of crossing the Atlantic broke a 150-year habit of staying home.

65. OCCUPATIONS — social stratification

758 relatives have occupation data (detailed occupation and cause-of-death lists are held in the separate story catalog and causes-of-death reports; this is the statistical-stratification summary).

Occupational categories and lifespan

Category	Count	Avg lifespan
Education/Church/Medicine	28	70.4y
Bauer/Farmer	65	67.5y
Handwerker (craftsman)	74	66.4y
Kaufmann/Trade	31	66.2y
Heuerling (tenant)	76	65.2y
Knecht/Laborer	91	59.7y
Kötter (cottager)	10	57.6y

A clear social gradient: the educated/clergy/medical class lived longest (70.4y), landowning farmers next (67.5y), while laborers and cottagers — the landless bottom of the rural hierarchy — died youngest (57–60y). The 10-13 year gap between the top and bottom of the social order mirrors the rural/urban and class divides seen throughout the tree.

The disappearance of the Heuerling system

Era	Farmer	Heuerling	Craftsman	Laborer
1750–1799	16%	35%	17%	8%
1800–1849	13%	22%	19%	16%
1850–1899	6%	2%	12%	24%
1900–1949	5%	0%	6%	8%

The Heuerling (landless tenant) class — 35% of recorded occupations in the late 18th century — collapses to near zero by 1850. This is the demographic engine of emigration: the Heuerlinge who could not inherit land and faced a shrinking tenant system are precisely the people who boarded ships for Ohio and Texas. The

occupational record and the emigration record describe the same social process from two angles. The "Laborer" category rising to 24% by 1850–99 reflects both the proletarianization at home and the American working class the emigrants joined.

ANALYSIS COMPLETE

This statistical analysis covers essentially every structured field in the GEDCOM: births, deaths, marriages, baptisms, witnesses, godparents, residences, occupations, relationship degrees, and shared ancestors. Together with the companion reports — the story catalog and the causes-of-death report — it forms a comprehensive three-part documentation of the 64,507-relative Kovermann family tree.

The unifying thread across all 65 sections: a sedentary, endogamous, Catholic-and-Protestant world of closed marriage circles in the Osnabrück and Minden-Lübbecke regions, transformed between 1838 and 1900 by confessionally-channeled emigration into a mobile American family centered on Putnam County, Ohio (Catholic) and Austin County, Texas (Protestant) — with one man, Johann Heinrich Recker gen Redeker, standing at the genealogical hinge connecting 28% of the entire tree.